ECEN 714: DIGITAL INTEGRATED CIRCUIT DESIGN
LAB REPORT - 9

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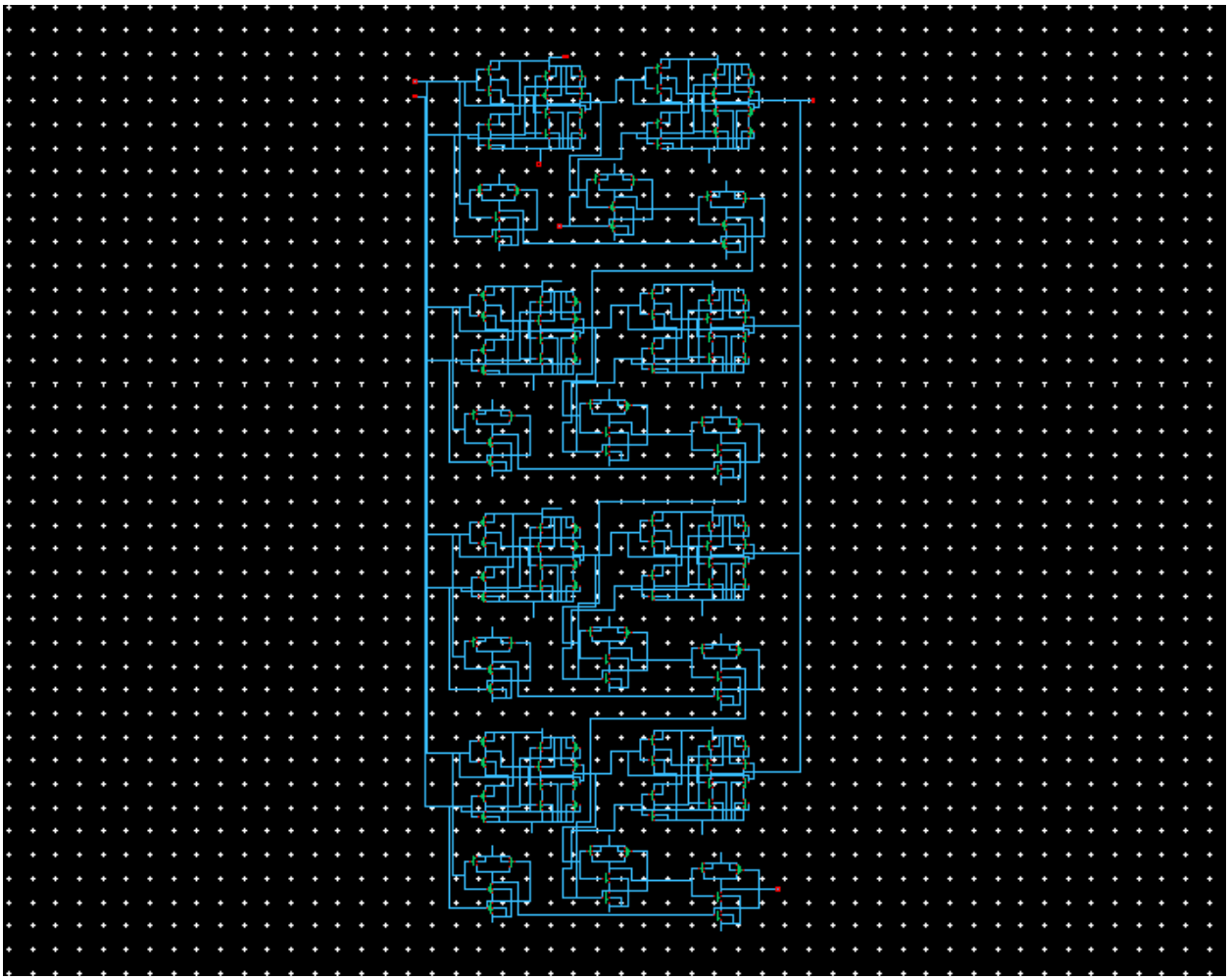
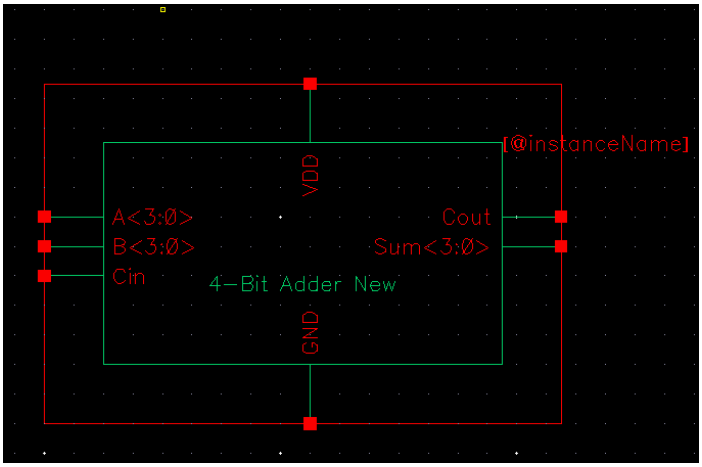
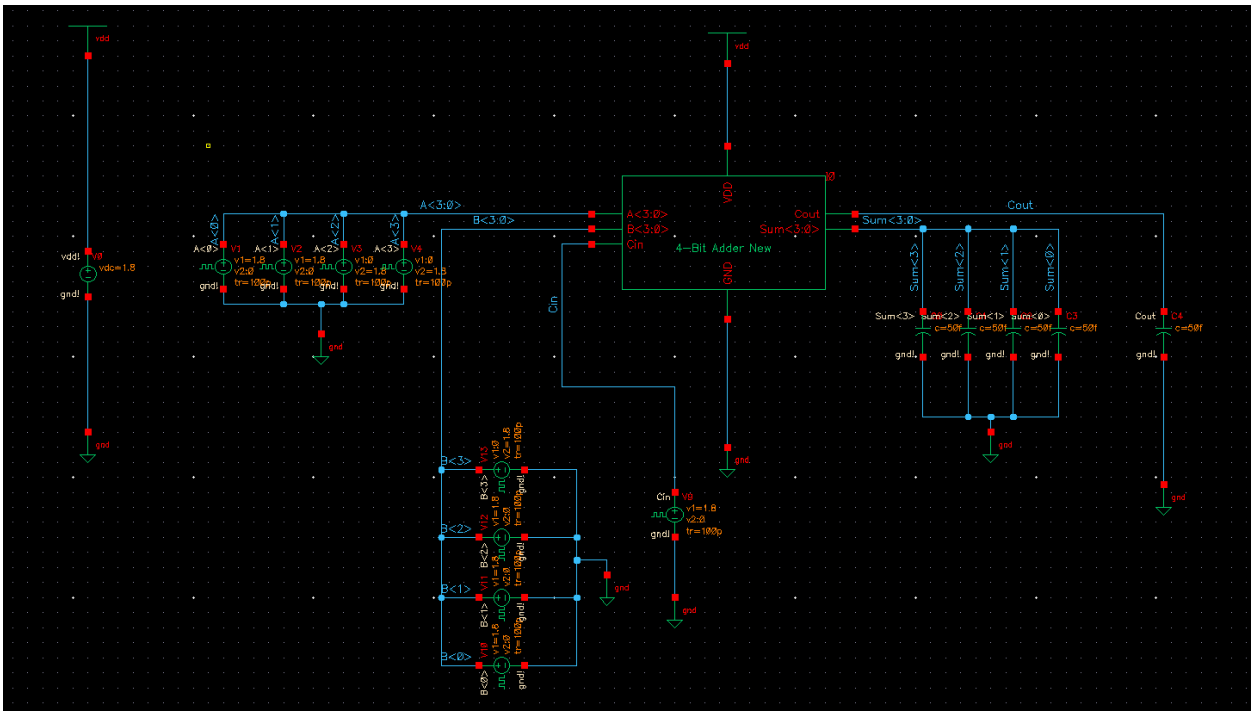
1. Schematic:



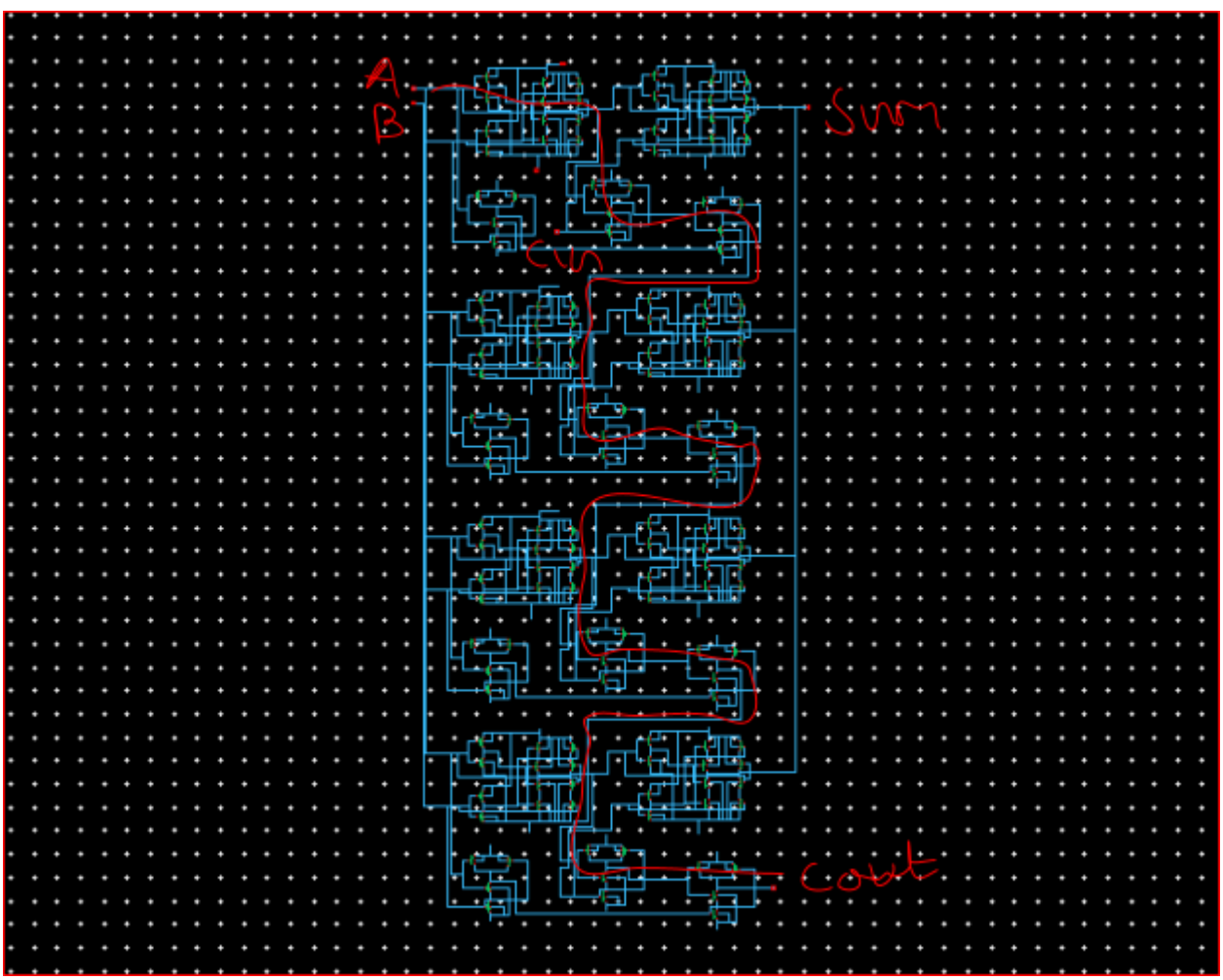
2. Symbol:



3. Simulation:



4. Critical Path:



5. GBH and Resizing Calculations:

1st Page:

$$G = \prod g_i \; ; (i = 1, 2, \ldots, 9)$$

$$= 4 \times \left(\frac{4}{3}\right)^8$$

$$= 39.98$$

$$H = C_{load} / C_{xor}$$

$$= \frac{30\,fF}{9.372\,fF} = \underline{\underline{3.2}}$$

$$B = \prod b_i \; ; (i = 1, 2, 3, \ldots, 9)$$

$$\Rightarrow b_1 = \frac{(C_{xor} + C_{nand2})}{C_{xor}}$$

$$= \frac{9.372 + 4.21}{9.372} = \underline{\underline{1.45}}$$

$$b_{2,4,6,8} = \frac{C_{xor} + C_{nand2}}{C_{nand2}}$$

$$= \frac{9.372 + 4.21}{4.21} = \underline{\underline{3.22}}$$

$$b_{3,5,7,9} = 1$$

$$\therefore B = 1.45 \times (3.22)^4 \times 1$$

$$= \underline{\underline{155.88}}$$

$$\therefore F = GBH = 155.88 \times 39.95 \times 3.2$$

$$= \underline{\underline{19{,}927.74}}$$

Resizing Calculations.

$$\hat{f} = F^{1/9} = 3$$

$$\because 2.4 < \hat{f} < 4$$

2nd Page:

NAND2 - 9

$$C_{in,9} = \frac{4/3 \times 30}{3} = 13.33 fF$$

$$\Rightarrow \text{Resize ratio} = \frac{C_{in,9}}{C_{gate}} = \frac{13.33}{2} = 6.66$$

$$W_p = 800 \times 6.66 = 5.328 \mu m$$

$$W_n = 400 \times 6.66 = 2.664 \mu m$$

NAND2 - 8

$$C_{in,8} = \frac{4/3 \times 13.33}{3} = 5.91 fF$$

$$\Rightarrow \text{Resize ratio} = \frac{C_{in,9}}{C_{gate}} = \frac{5.91}{2} = 2.95$$

$$W_p = 800 \times 2.95 = 2.36 \mu m$$

$$W_n = 400 \times 2.95 = 1.18 \mu m$$

NAND2 - 7

$$C_{in,7} = \frac{4/3 \times 15.28}{3} = 6.77 fF$$

$$C_{out,7} = C_{in,8} + C_{in,xor} = 5.91 + 9.372 = 15.28 fF$$

$$\Rightarrow \text{Resize ratio} = \frac{C_{in,7}}{C_{gate}} = \frac{6.77}{2} = 3.38$$

$$W_p = 800 \times 3.38 = 2.704 \mu m$$

$$W_n = 400 \times 3.38 = 1.352 \mu m$$

NAND2 - 6

$$C_{in,6} = \frac{4/3 \times 6.77}{3} = 3 fF$$

$$\Rightarrow \text{Resize ratio} = \frac{C_{in,9}}{C_{gate}} = \frac{3}{2} = 1.5$$

$$W_p = 800 \times 1.5 = 1200 nm$$

$$W_n = 400 \times 1.5 = 600 nm$$

3rd Page:

NAND2 - 5

$$C_{in,5} = \frac{4/3 \times 12.37}{3} = \underline{\underline{5.48}}\ fF$$

$$C_{out,5} = C_{in,6} + C_{in,xor} = 9.372 + 3 = 12.372\ fF$$

$$\Rightarrow \text{Resize ratio} = \frac{C_{in,5}}{C_{gate}} = \frac{5.48}{2} = 2.74$$

$$W_p = 800 \times 2.74 = 2.192\ \mu m$$

$$W_n = 400 \times 2.74 = 1.096\ \mu m$$

NAND2 - 4

$$C_{in,4} = \frac{4/3 \times 5.48}{3} = \underline{\underline{2.42}}\ fF$$

$$\Rightarrow \text{Resize ratio} = \frac{C_{in,4}}{C_{gate}} = \frac{2.42}{2} = 1.21$$

Since, $1.21 < 2$, no resize is needed.

NAND2 - 3

$$C_{in,3} = \frac{4/3 \times 12.11}{3} = \underline{\underline{5.368}}\ fF$$

$$C_{out,3} = C_{in,4} + C_{xor} = 2.74 + 9.372 = 12.112\ fF$$

$$\Rightarrow \text{Resize ratio} = \frac{C_{in,}}{C_{gate}} = \frac{5.368}{2} = 2.68$$

$$W_p = 800 \times 2.68 = 2.144\ \mu m$$

$$W_n = 400 \times 2.68 = 1.072\ \mu m$$

NAND2 - 2

$$C_{in,2} = \frac{4/3 \times 5.36}{3} = \underline{\underline{2.37}}\ fF$$

$$\Rightarrow \text{Resize ratio} = \frac{C_{in,}}{C_{gate}} = \frac{2.37}{2} = 1.18$$

$$W_p = 800 \times 1.18 = 951\ nm$$

$$W_n = 400 \times 1.18 = 475\ nm$$

XOR - ①

$$C_{in,1} = \frac{4 \times 11.74}{3} = \underline{\underline{15.61}}\ fF$$

$$\begin{cases} C_{out,1} = C_{in,2} + C_{XOR} \\ \quad = 2.37 + 9.372 \\ \quad = 11.742 \end{cases}$$

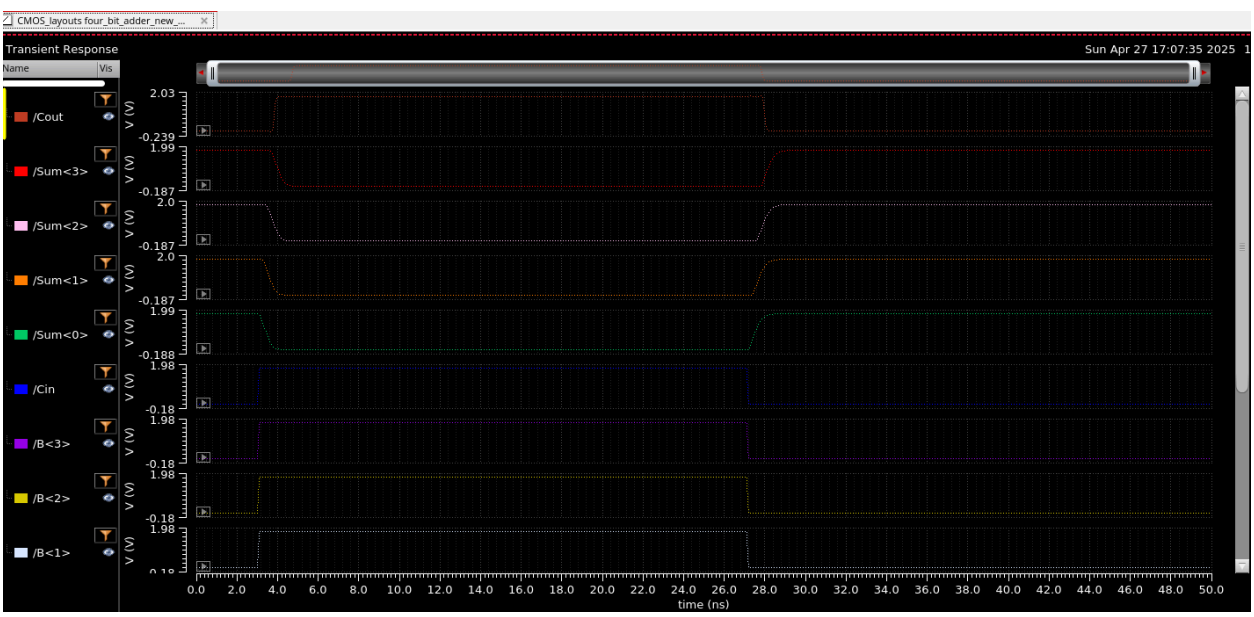
$$\Rightarrow \text{Resize ratio} = \frac{C_{in,1}}{C_{gate}} = \frac{15.61}{9.372} = 1.66$$

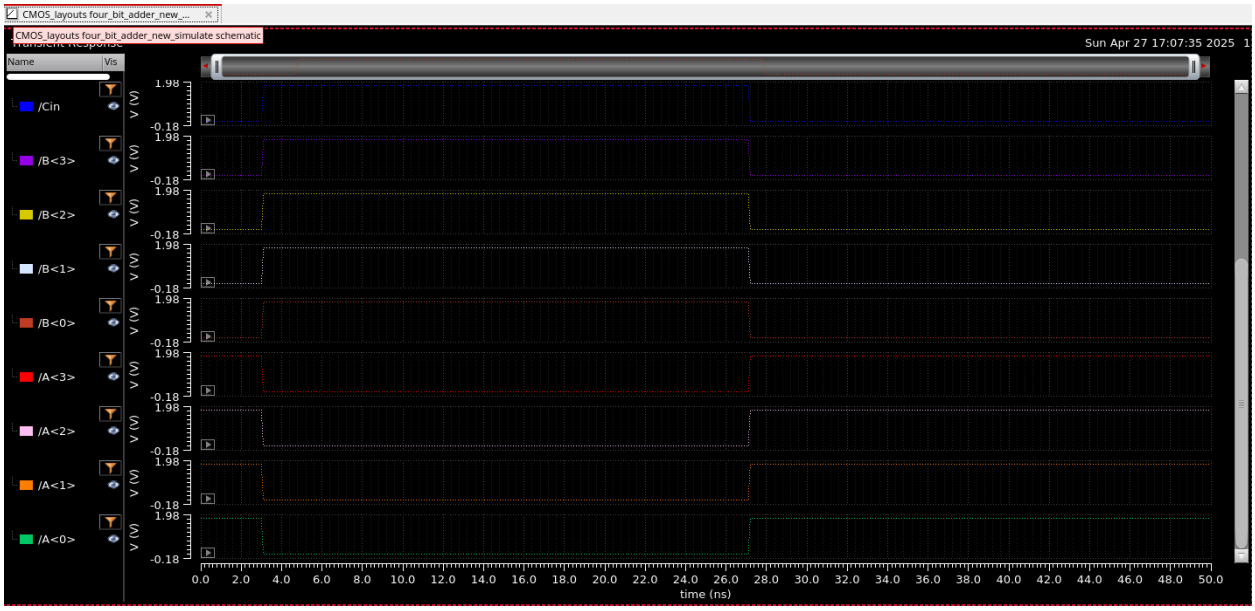
$$W_p = 1400 \times 1.66 = 2.3\ \mu m$$

$$W_n = 400 \times 1.66 = 664\ nm$$

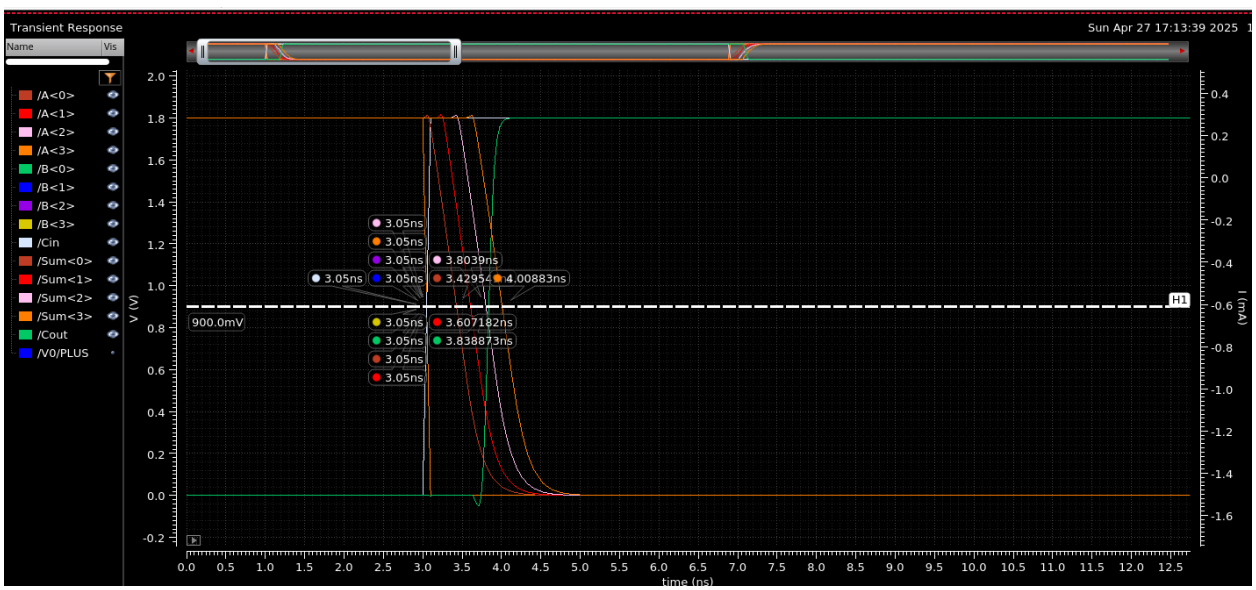
6. Simulation:

Case 1: A = 0000, B = 1111, Cin = 1

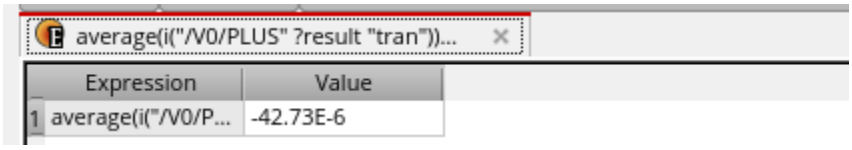




For Delay Calculations:

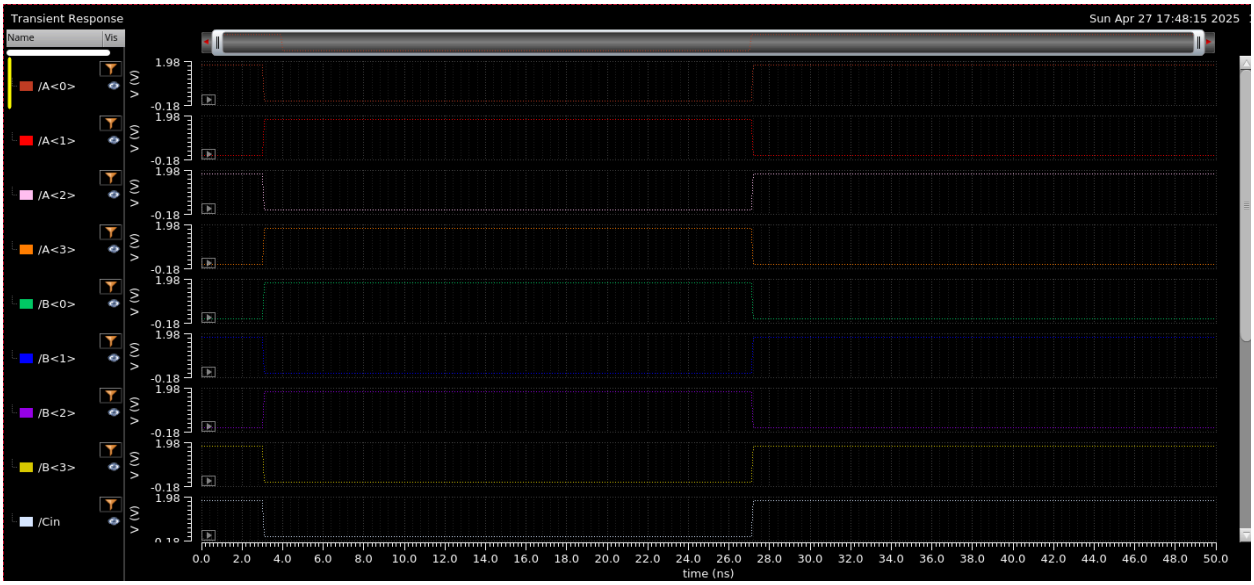


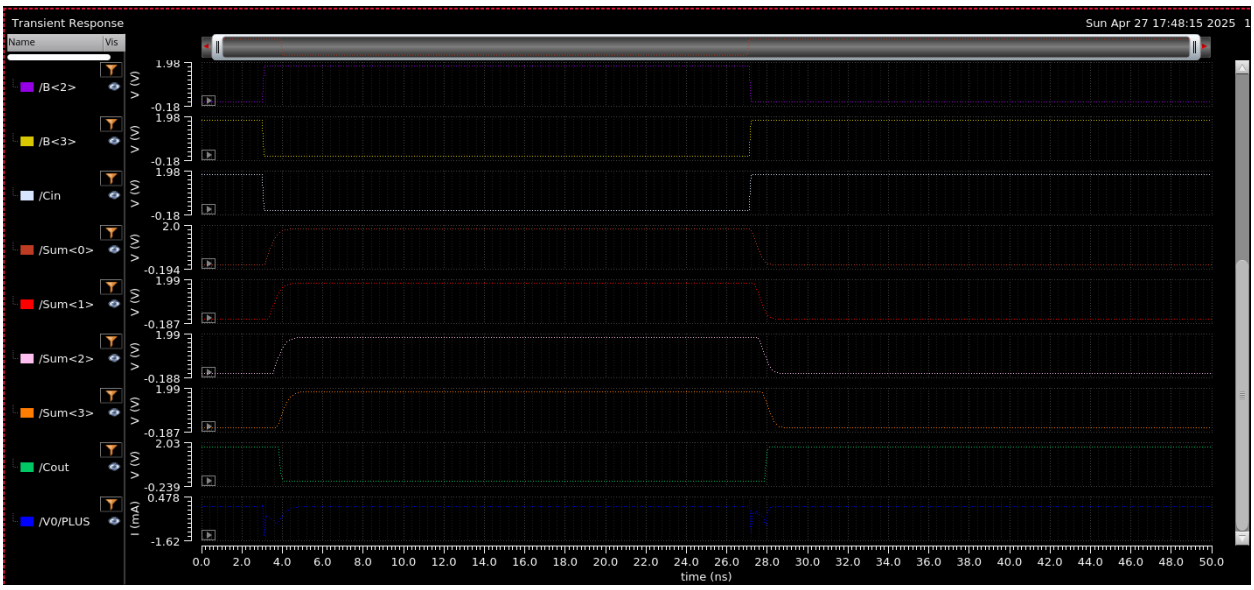
Power from VDD:



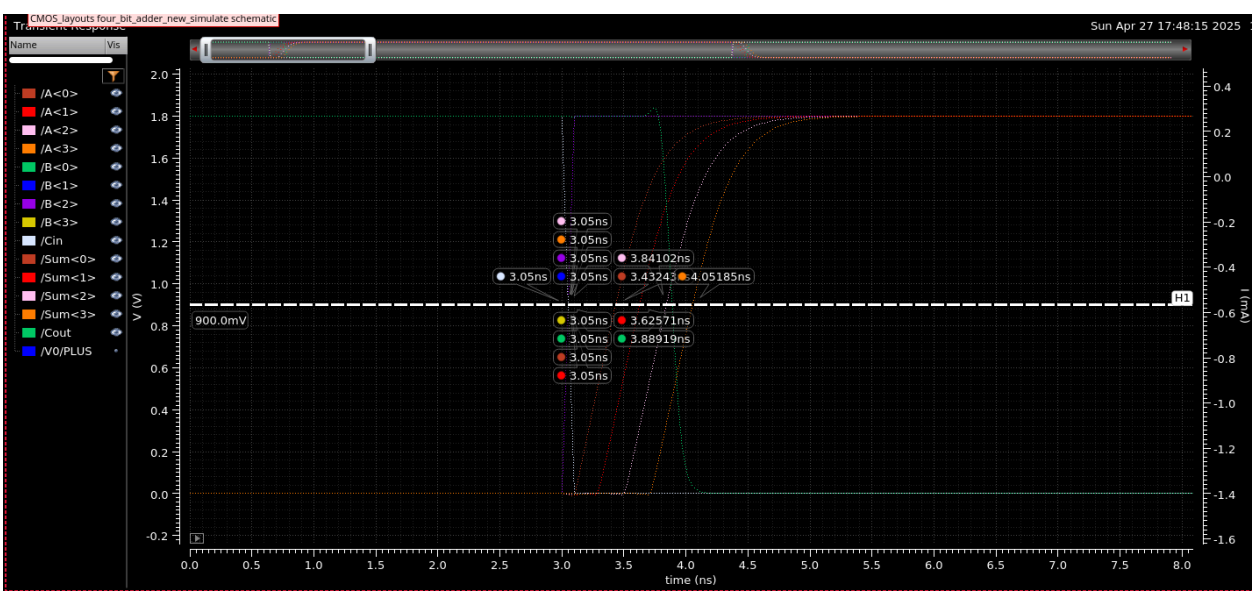
Power = 42.73 uW

Case 2: A = 1010, B = 0101, Cin = 0

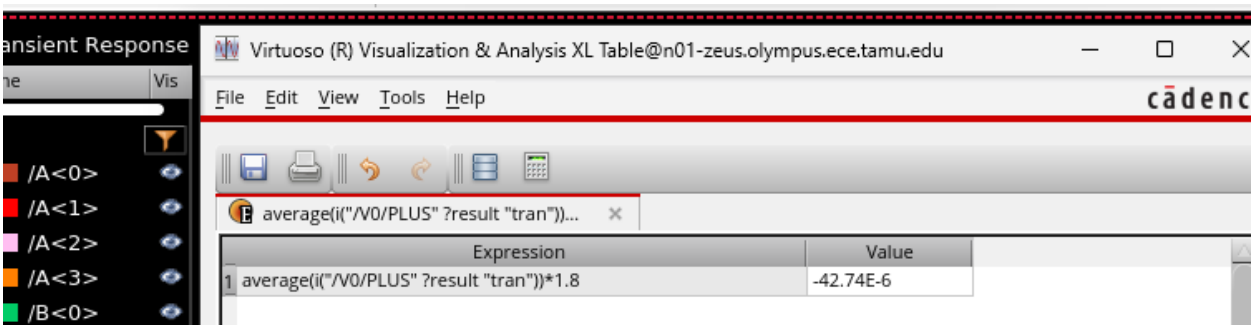




For Delay Calculations:

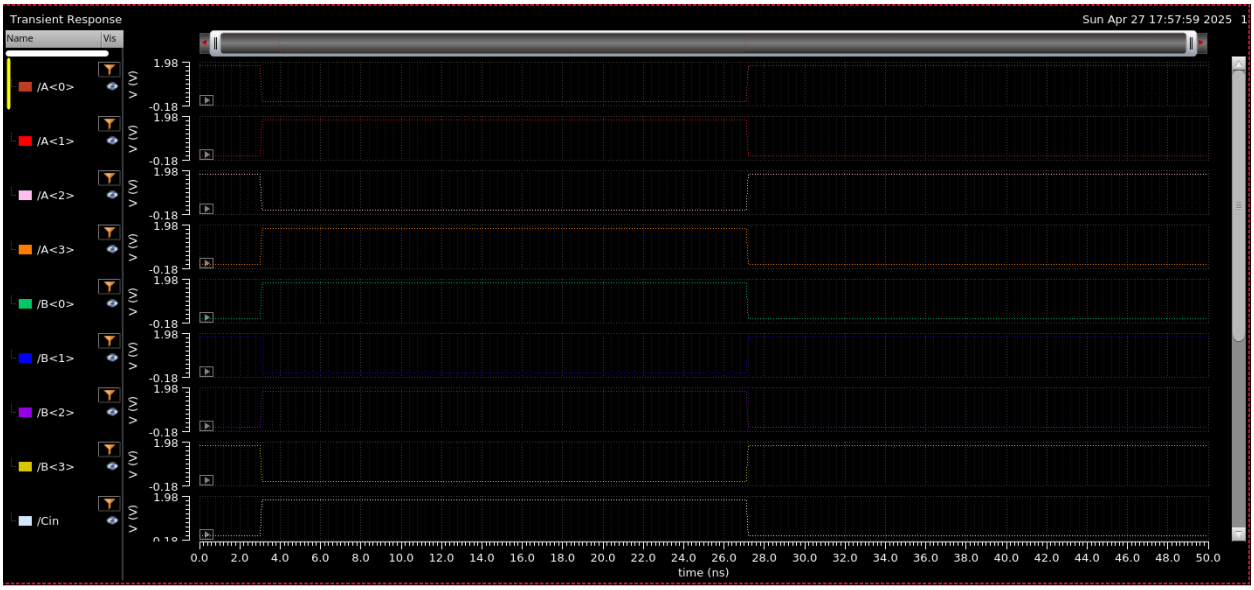


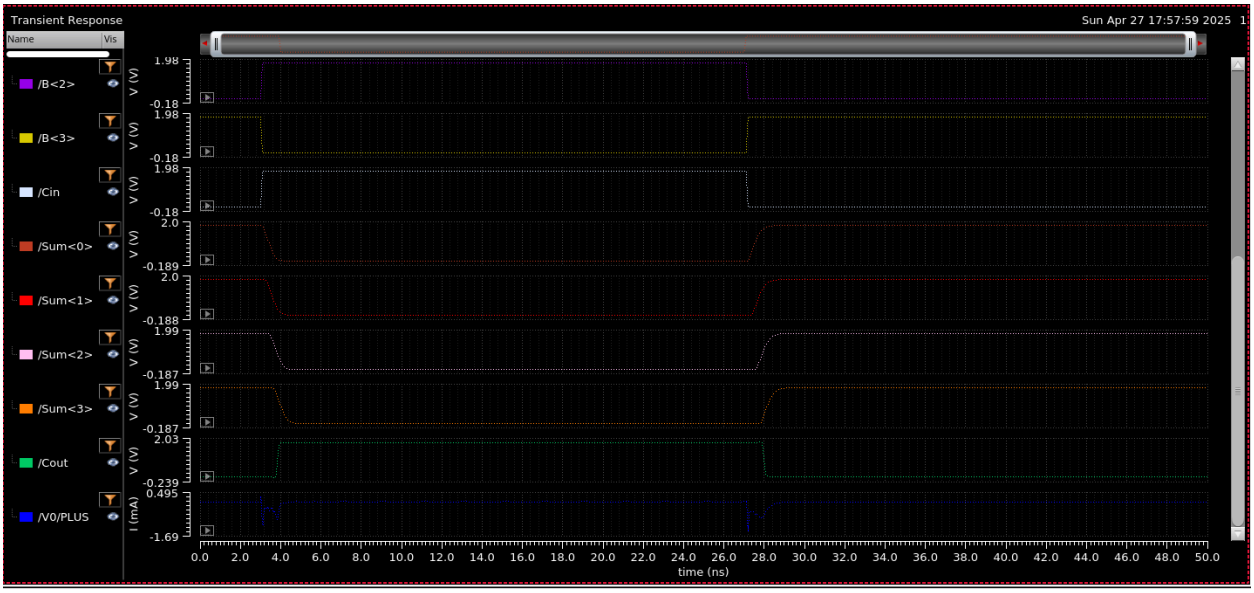
Power from VDD:



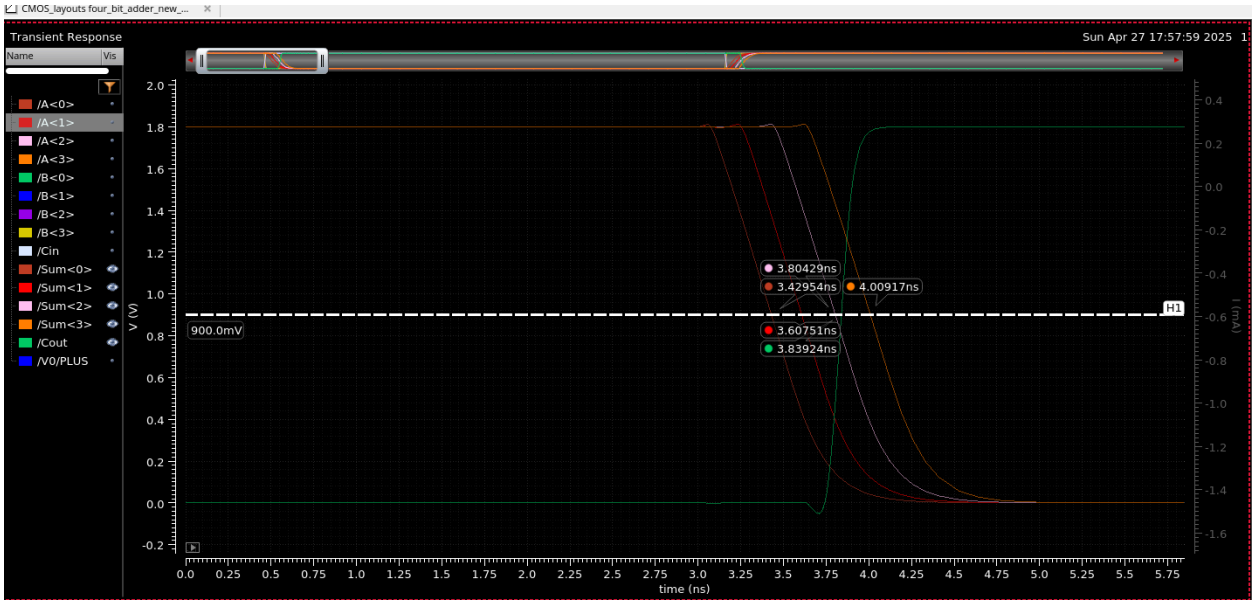
Power = 42.74 uW

Case 3: A = 1010, B = 0101, Cin = 1

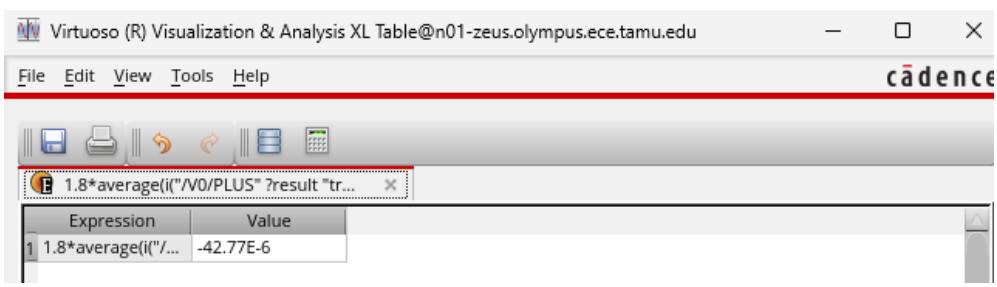




For Delay Calculations:

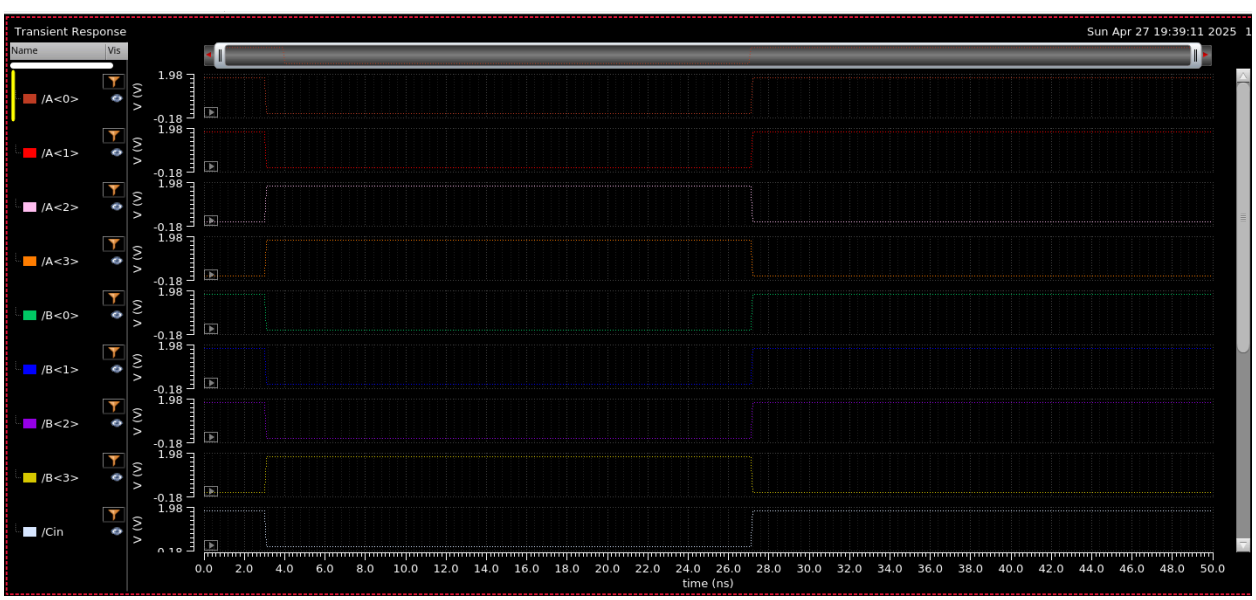


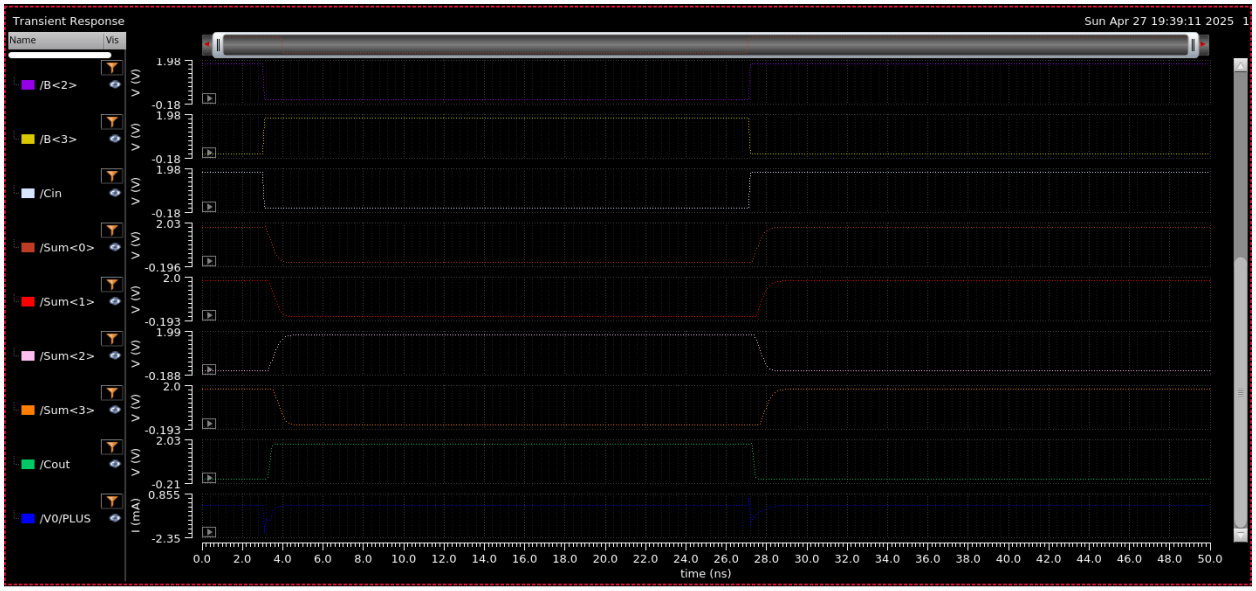
For VDD Power Consumption:



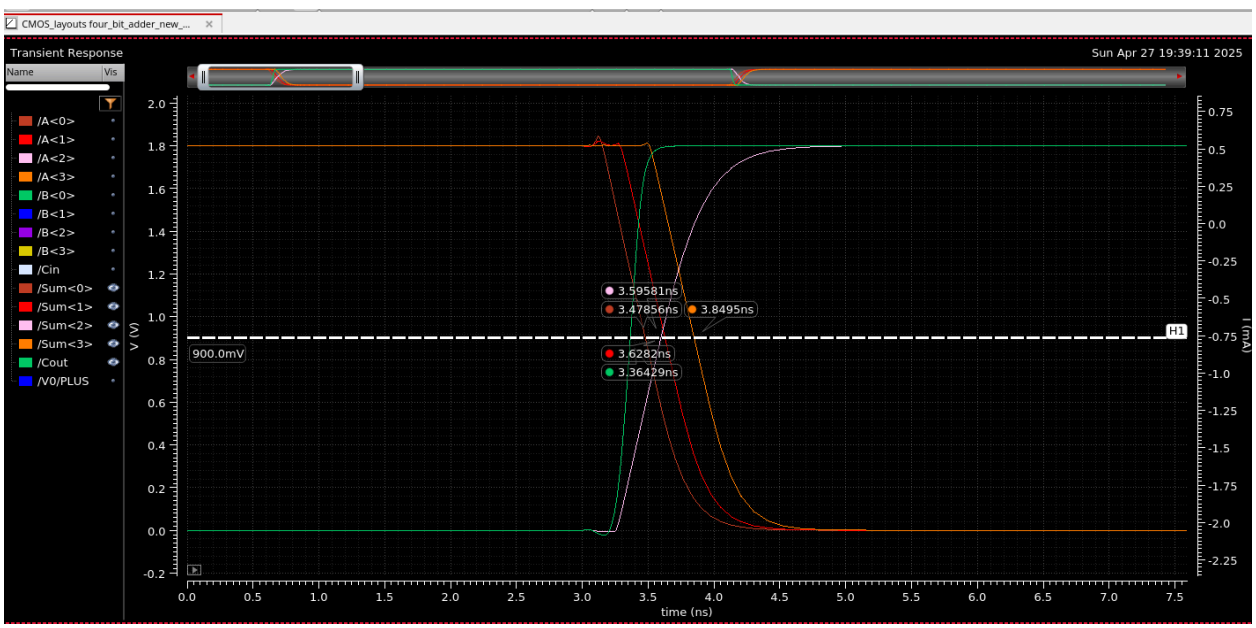
Power = 42.77 uW

Case 4: A = 1100, B = 1000, Cin = 0

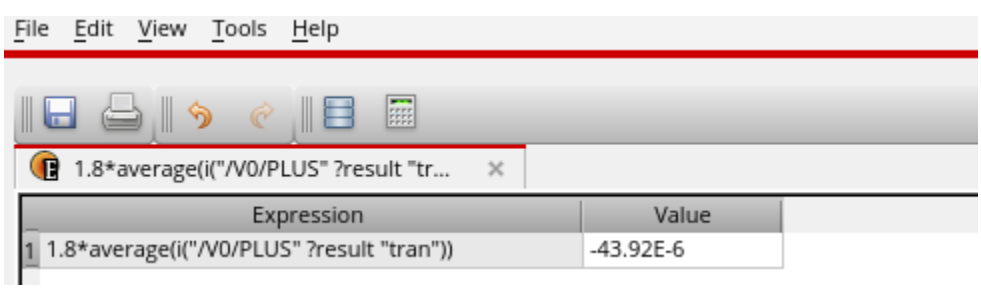




For Delay Calculations:



For VDD Power Consumption:



Power = 43.92 uW

Delay Table:

CASE	PIN	NON-OPTIMISED (ns)	OPTIMISED (ns)
A = 0000 **B = 1111** **CARRY IN = 1**	SUM0	0.371	0.379
	SUM1	0.648	0.557
	SUM2	0.915	0.753
	SUM3	1.199	0.95
	Cout	1.234	0.788
A = 1010 **B = 0101** **CARRY IN = 0**	SUM0	0.373	0.382
	SUM1	0.666	0.575
	SUM2	0.942	0.791
	SUM3	1.236	1.001
	Cout	1.24	0.839
A = 1010 **B = 0101** **CARRY IN = 1**	SUM0	0.371	0.379
	SUM1	0.648	0.557
	SUM2	0.915	0.754
	SUM3	1.199	0.959
	Cout	1.209	0.789
A = 1100 **B = 1000** **CARRY IN = 0**	SUM0	0.42	0.428
	SUM1	0.674	0.578
	SUM2	0.611	0.545
	SUM3	0.947	0.799
	Cout	0.369	0.314

As we can see the Cout delay is significantly reduced.

Power Comparison Table:

Cases**	**Optimised (uW)**	**Non-Optimised (uW)**
Case 1	42.73	41.068
Case 2	42.74	43.15
Case 3	42.77	41.01
Case 4	43.92	42.02

** According to the cases given above in the simulation pictures.

Area Comparison Table:

Optimised (u^2m^2)	**Non-Optimised (u^2m^2)**
28.8	16.8